

Esercizi

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1. *Risolvere le seguenti equazioni pure:*

(a) $(x-1)^2 + (x+1)^2 = (x+1)(x-1) + 5$	(d) $(x-5)^2 = 25$
(b) $(3+x)(x-3) - 2(x+1)^2 = 2 - (x+2)^2 + x^2$	(e) $(2x-3)^3 - (2x-3)(4x^2-6x+9) = 18x-3$
(c) $\frac{x^2+3}{4} = \frac{(x+3)^2 + (x-3)^2}{2} - 9$	(f) $(1-2x)^2(1+2x)^2 - (4x^2-3)(4x^2+3) = 0$

2. *Risolvere le seguenti equazioni spurie:*

(a) $(2x-1)(2x+3) = -3$	(c) $x^2 - \frac{1}{2}(x+1)^2 = 1 - \frac{x+3}{2}$
(b) $(x+1)^3 - (x+2)^3 = -7$	(d) $(x+1)(x+3) + (x-1)^2 = (x+2)^2$

3. *Risolvere le seguenti equazioni incomplete:*

(a) $x^2 + (x-2)^2 = (x+2)^2$	(e) $(2x-1)(2x+1) = (3x-1)(3x+1)$
(b) $x^2 + (x-2)^2 = 1 - (x+2)^2$	(f) $(2x-1)(3x-2) = 5x+2$
(c) $(x+2)(x-2) + (x-1)^2 = x-3$	(g) $\frac{(x-1)^2}{2} - \frac{x^2+1}{3} = \frac{1}{6}$
(d) $(x+1)^3 - (x-1)^3 = x+2$	(h) $\left(2x - \frac{1}{2}\right)^2 = \left(3x - \frac{1}{3}\right)^2$

4. *Risolvere le seguenti equazioni complete:*

(a) $x^2 - x - 30 = 0$	(g) $(x+1)^2 = (2x-3)^2$
(b) $3x^2 - 2x - 1 = 0$	(h) $(2x-1)(2x+1) + (3x-1)(2x-3) = 6x-1$
(c) $-x^2 = x - 12$	(i) $\frac{x+1}{2} + \frac{(x-1)^2}{3} + \frac{5}{6} = (x+2)(x-3) + 6$
(d) $2x^2 = (x+2)(x-3) + 12$	(j) $\frac{1}{2}(x+1)(x-1) - \frac{2}{3}(x+2) = 2 - (x+1)^2$
(e) $(x-3)^2 + (5-x)^2 = 2x-6$	(k) $2(2x+1)^2 - (2x+1) = 4x^2 - 1$
(f) $(2x-1)^2 = 3x(x-2)$	(l) $(x-2)^2(x+1) = (x-2)(x+1)^2$